

Day 27 - OSPF (pt 2)

Sunday, September 6, 2026

10:46 AM

OSPF Information

OSPF Metric/Cost

OSPF's metric is called cost.

- It is calculated by dividing a **reference bandwidth** value by the interface's **bandwidth**.
 - The default **reference bandwidth** is **100 mbps**
 - All values **less** than **1** (**0.01** for example) will be converted to **1**
 - The formula to calculate **cost** is **reference bandwidth/interface bandwidth**
 - The **OSPF cost** to a destination is the total **cost** of the **outgoing** interfaces

Loopback Interface Cost

Loopback interfaces have an **outgoing** cost of **1**.

Reference bandwidth configuration

You should configure a **reference bandwidth** greater than the fastest links in your network, to allow for more accurate costs.

All OSPF routers should have the same **reference bandwidth**.

The command to change the **reference bandwidth** is: **auto-cost reference-bandwidth MEGABITS-PER-SECOND**

Speed VS. Bandwidth

By default, the **bandwidth** matches the interface **speed**. Changing the **bandwidth** doesn't change the **speed** at which the interface runs at, instead, it changes the way that multiple calculations are done (OSPF cost, EIGRP metric, etc.)

If you want to change the **cost** value for OSPF, It is recommended to change it through the **ip ospf cost** **COST** command.

Methods to Change OSPF Cost

1. **Reference bandwidth**
 - auto-cost reference-bandwidth **megabits-per-second**
2. **Manual Configuration**
 - ip ospf cost **cost**
3. **Interface bandwidth**
 - bandwidth **kilobits-per-second**

Extra OSPF Configuration

- You can activate **OSPF** directly on an interface with this command:
 - **ip ospf PROCESS-ID area AREA**
- You can configure all interfaces as **OSPF** passive interfaces by default with this command:
 - **passive-interface default**
- You can change the **reference bandwidth** with this command:
 - **auto-cost reference-bandwidth MEGABITS-PER-SECOND**
- You can change the **cost** or **bandwidth** value for a specific interface with this command:
 - **ip ospf cost COST**

"Demons In Texas Eat Eels Like Fries"

OSPF Neighbors

Once routers become OSPF neighbors, they automatically start sharing network information, calculating routes and etc.

- When **OSPF** is activated on an interface, the router starts sending **OSPF Hello** messages out of the interface at intervals determined by the **Hello timer**. These are used to discover potential **OSPF** neighbors.
 - The default hello timer is 10 seconds on an **Ethernet** connection.
 - **Hello** messages are **multicast** to 224.0.0.5 (All **OSPF** routers).

- **OSPF** messages are encapsulated in an IP header, with a value of **89** in the protocol field.
- Many different messages are sent out in the **OSPF** neighbor process. You can find the message types summarized [here](#).

State 1: Down

- When an interface is enabled, the first router sends an **OSPF Hello** message with its own **Router ID** and a **neighbor Router ID** with a value of **0.0.0.0**, as it does not know any neighbor's actual **Router ID**.
- In this stage, the **OSPF** neighbor state is currently **Down**.

State 2: Init

- The neighbor router receives the first router's **Hello** message, and adds the first router's **Router ID** into its **OSPF** neighbor table.
- For the neighbor, its relationship with the first router is currently in the **Init** state.

State 3: Two-way

- The neighbor router will send a **Hello** message containing both **Router IDs**.
- The first router will accept the message and add the neighbor into its **OSPF** neighbor table in the **2-way** state.
- After that, the first router will send a **Hello** message back, the neighbor will accept it and add the first router to its **OSPF** neighbor table in the **2-way** state as well.
- If both routers are in the **2-way** state, that means they have successfully become **OSPF** neighbors and are ready to share **LSAs** in order to build a common **LSDB**.
- In some network types, a **DR (Designated Router)** and a **BDR (Backup Designated Router)** will be elected at this point.

[What does the 2-way state mean exactly?](#)

The 2-way state means the router has received a **Hello** packet with its own **Router ID** in it.

State 4: Exstart

In the Exstart state, the routers will decide which router will be in charge of starting the exchange.

- The router with the **higher Router ID** will become the **Master** and the router with the **lower Router ID** will become the **Slave**.
- To decide the **Master** and **Slave**, they exchange **DBD (Database Description)** packets.

State 5: Exchange

In the Exchange state, the routers will exchange DBDs which contain a list of the LSAs in their LSDBs.

- The **DBDs** do not contain detailed information, just basic information about the **LSAs** present in the database.
- The routers compare the information in the **DBD** they received to the information in their own **LSDB** to determine which **LSAs** they must receive from their neighbors.

State 6: Loading

In the Loading state, routers request the LSAs they need through Link State Request (LSR) messages.

- **LSAs** are sent in **LSU** messages.
- The routers send **Link State Acknowledgement (LSAck)** to their neighbors to acknowledge that they have received the missing **LSAs**.

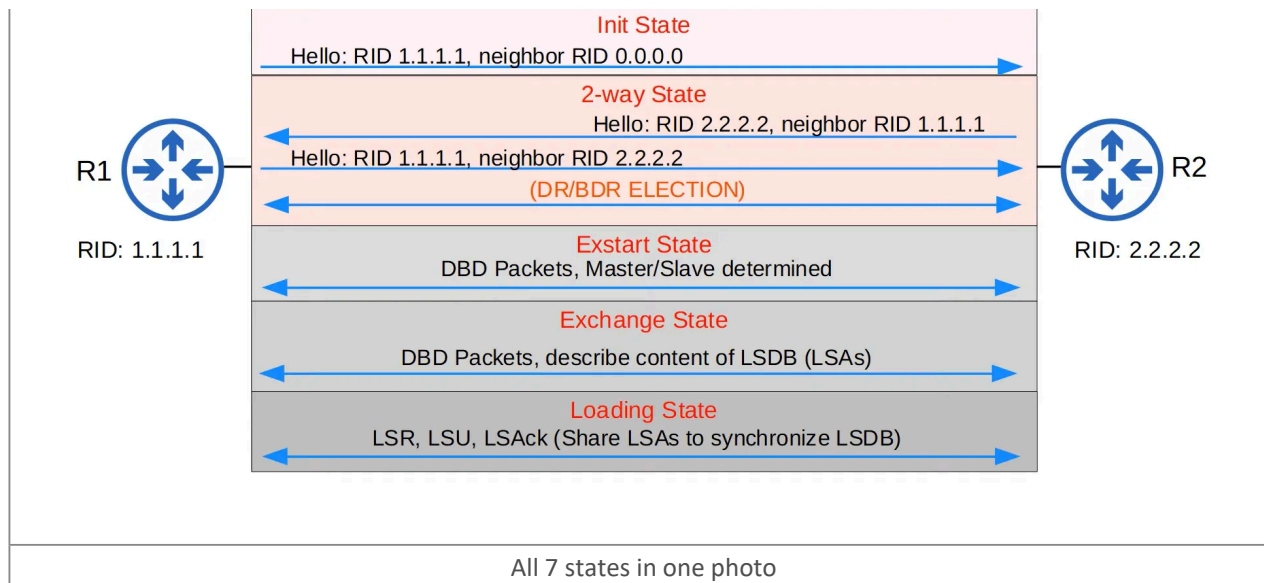
State 7: Full

In the Full state, the routers have reached full OSPF adjacency and have identical LSDBs.

- The routers will continue to share **LSAs** when network changes happen, to make sure that every router has the same map of the network. (Same **LSDBs**)
- They do that by sending and listening for **Hello** messages (Default **Hello** timer is **10 seconds** on **Ethernet**) to maintain the adjacency.
- Every time a **Hello** message is received, the **Dead** timer is reset (Default **Dead** timer is **40 seconds** on **Ethernet**).
 - If the **Dead** timer reaches **0** and no **Hello** message is received, the neighbor is removed.

Summary Photo





Acronym

A good acronym that helped me memorize the different OSPF states was: "Demons In Texas Eat Eels Like Fries"

OSPF Message Types Table

Type	Name	Purpose
1	Hello	Neighbor discovery and maintenance.
2	Database Description (DBD)	Summary of the LSDB of the router. Used to check if the LSDB of each router is the same.
3	Link State Request (LSR)	Requests specific LSAs from the neighbor.
4	Link State Update (LSU)	Sends specific LSAs to the neighbor.
5	Link State Acknowledgement (LSAck)	Used to acknowledge that the router received a message.